

DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

PLATFORM, ELECTRICAL & INTERFACES

Coupled-machine platform role, electrical interface categories, safety boundaries, verification, and public release constraints.

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.6 | 10 AUGUST 2026

Prepared by	Herder Elektronische Systemen
Document scope	Coupled-machine platform role, electrical interface categories, safety boundaries, verification, and public release constraints.
Evidence basis	Controlled internal platform inspection and bench demonstration summarized at black-box level; no manufacturing or setup detail.
Distribution	Public customer-facing release; implementation IP, source artifacts, setup procedures, and reproducible design detail excluded.
DOCUMENT PURPOSE Defines the public functional, interface, safety, verification, and acceptance requirements for the physical dynamometer platform while withholding reproducible electrical and mechanical design information.	

Functional Requirements Document Framework

WHY THIS IS AN FRD
This document defines the required external behavior of the platform and its interface categories, together with verification and acceptance criteria. It is intentionally not a schematic package, manufacturing specification, assembly guide, or setup procedure.

Purpose, Scope, and System Boundary

Coupled-machine platform role, electrical interface categories, safety boundaries, verification, and public release constraints.

Boundary element	Included in this FRD	Explicitly excluded
Mechanical platform	Coupled dual-motor functional role and externally observable behavior	Dimensions, tolerances, assembly drawings, fixtures, fabrication details
Electrical platform	Controller/inverter roles and interface categories	Schematics, BOMs, connector/pin maps, harnesses, component values, board files
Commissioning	Requirement that the system can be safely commissioned internally	Setup steps, wiring checks, configuration procedures, calibration instructions

Users and Operational Scenarios

Actor	Primary interaction	Expected outcome
Operator	Use the assembled platform under approved internal procedures	Controlled access to drive/load operation
Test engineer	Connect approved instrumentation and execute controlled tests	Reviewable evidence without public manufacturing detail
Customer / reviewer	Assess platform function, interfaces, status, and limitations	Understand capability without receiving reproducible design information

Functional Requirements

ID	The system shall...	Rationale / explanation	Verification
PLAT-FR-001	provide two	Defines the primary	Inspection +

ID	The system shall...	Rationale / explanation	Verification
	mechanically coupled motor positions suitable for drive/load dynamometer operation.	physical function.	demonstration
PLAT-FR-002	provide independent power-conversion channels for the two machines.	Allows separate drive and load commands.	Inspection + functional demonstration
PLAT-FR-003	provide independent machine feedback paths to the embedded control system.	Supports closed-loop control and machine-level observability.	Functional demonstration
PLAT-FR-004	provide a supervisory communications interface compatible with the control subsystem.	Allows external command and diagnostic integration.	Interface demonstration
PLAT-FR-005	support controlled bench characterization across the demonstrated operating envelope.	Defines platform suitability without disclosing test procedure.	Bench demonstration
PLAT-FR-006	prevent the public release from disclosing manufacturing, wiring, commissioning, or reproducible assembly information.	Maintains the IP and ownership boundary.	Release inspection

Performance and Quality Requirements

ID	Requirement	Acceptance basis
PLAT-NFR-001	The platform shall be treated as an engineering bench asset rather than a production-qualified product.	Public documentation contains explicit qualification limitations.
PLAT-NFR-002	Public documentation shall identify third-party hardware at product-family level only where	No vendor CAD, fabrication, licensed package, or detailed schematic is redistributed.

ID	Requirement	Acceptance basis
	useful to describe system context.	
PLAT-NFR-003	The public release shall contain no detailed pinouts, connector maps, harness drawings, BOMs, schematics, board files, fabrication outputs, or setup procedures.	Package inspection confirms absence.
PLAT-NFR-004	Any operational evidence shown publicly shall avoid institution-specific markings, private identifiers, or unpublished laboratory information.	Media/document review confirms sanitization.

External Interface Requirements

Interface	Required behavior	Failure / boundary condition
Power	Supports controller and dual inverter operation from an appropriate controlled bench source.	No voltage wiring sequence, polarity procedure, fuse sizing, or connector detail is published.
Motor actuation	Two independent three-phase machine interfaces support drive/load operation.	No phase-to-pin or harness mapping is published.
Feedback	Two independent feedback channels support machine control and monitoring.	No electrical pinout, level, scaling, or decoder configuration is published.
Communications	External supervisory communications support command and diagnostic integration.	No bus identifiers, payloads, termination instructions, or wiring diagrams are published.

1. Platform Role and Delivered Capability

The platform is a coupled dual-motor bench asset built around a commercially available embedded controller and inverter stages. This FRD defines only the externally relevant platform functions and interfaces. It intentionally does not publish enough electrical, mechanical, or connector detail to reproduce or commission the hardware.

- Two permanent-magnet synchronous machines are mechanically coupled for drive/load operation.
- Commercial controller and inverter hardware provide the embedded power-electronics foundation.
- The platform exposes power, motor, feedback, communications, and service interfaces at a functional level.
- Detailed wiring, pinouts, schematics, BOMs, board files, cable construction, and assembly procedures are excluded.

2. Platform Functional Boundary

The platform requirements focus on mechanical coupling, independent machine actuation, feedback availability, and safe bench integration. Detailed physical implementation remains controlled.

Capability	Public requirement / acceptance basis	Status
Coupled shaft	Both machines operate on a common mechanical load path suitable for dynamometer testing.	Demonstrated
Independent actuation	Each machine is driven by its own inverter stage under embedded control.	Demonstrated
Independent feedback	Each machine provides a distinct feedback path to the control system.	Demonstrated
Supervisory integration	The platform supports an external engineering host for commands and diagnostics.	Demonstrated

3. Electrical and Physical Interface Boundary

Only interface categories and required behavior are public. Connector numbers, board revisions, exact pins, wire colors, harness construction, termination details, protection-component values, and commissioning sequences are intentionally omitted.

Interface category	Required behavior	Public boundary
DC power	Provide the controlled energy source required by the embedded controller and inverter stages.	No wiring or commissioning instructions
Motor power	Provide independent three-phase actuation paths for both machines.	No phase mapping or connector detail
Position/speed feedback	Maintain separate feedback channels for each machine.	No pinout, electrical-level, or decoder configuration

Interface category	Required behavior	Public boundary
Supervisory communications	Provide a compatible external control/diagnostic path.	No physical-layer construction or protocol detail

Selected Public Hardware Evidence

The photographs below show the commercial hardware set and bench-level electrical integration used for the platform. They are included as visual evidence only. They do not provide wiring instructions, pin mappings, assembly procedures, calibration values, or manufacturing information.

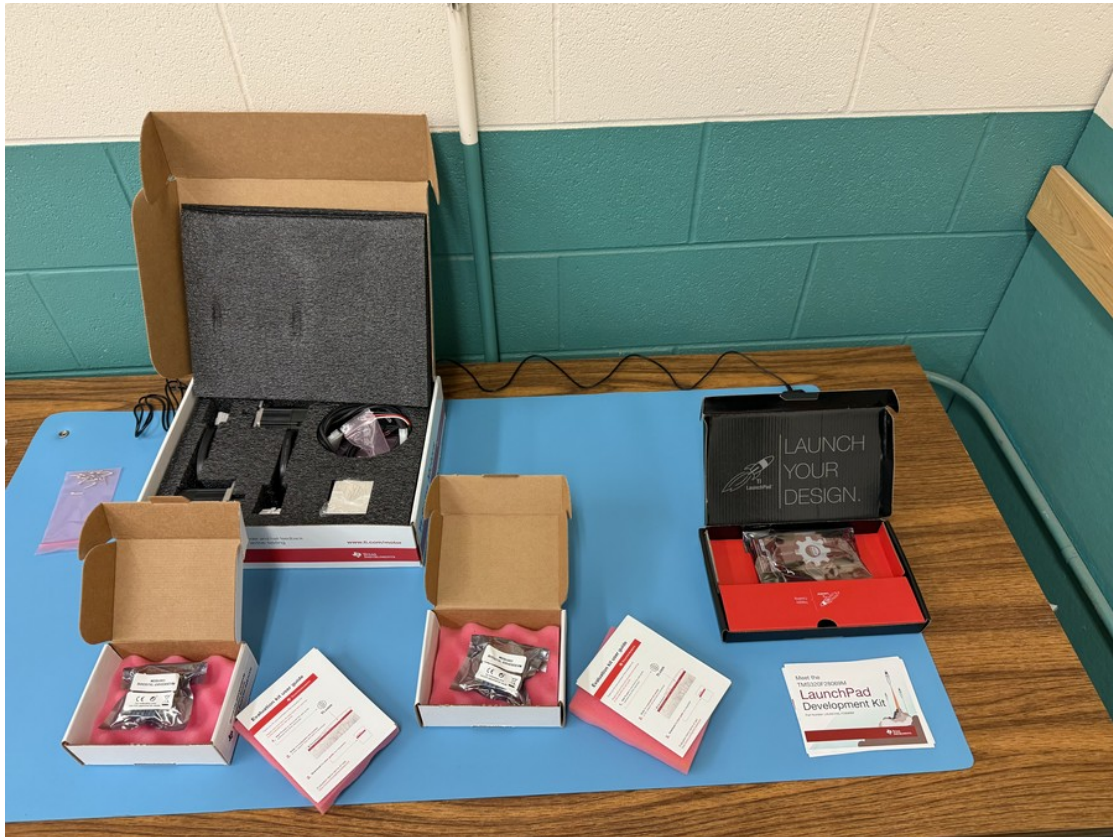


Figure 1. Public hardware overview showing the motor-control equipment set used for bench integration.



Figure 2. Motor and cabling hardware retained as non-sensitive physical evidence.

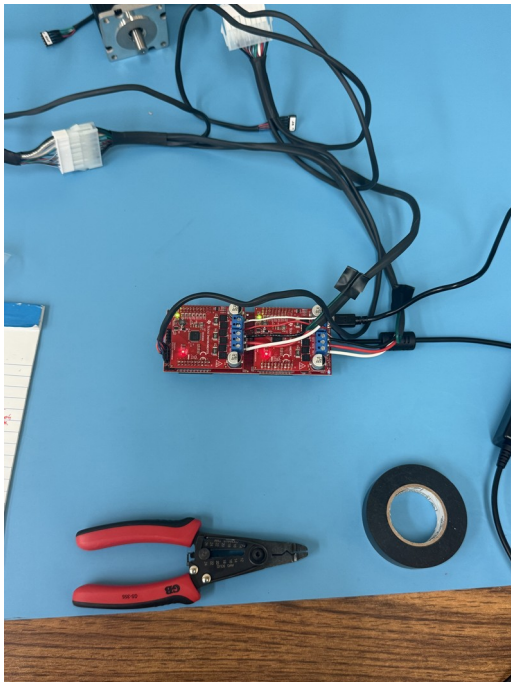


Figure 3. Controller/inverter bench stack shown without published wiring or connector instructions.

4. Safety, Ownership, and Qualification

The public platform description is intentionally non-reproducible. It demonstrates engineering scope without distributing manufacturing information or institution-specific laboratory material.

- Commercial TI hardware remains subject to the manufacturer’s documentation and licensing.
- Physical test equipment ownership is separate from authorship of the public engineering documentation.
- No production safety, EMC, environmental, enclosure, ingress, endurance, or metrology certification is claimed.

Requirements Verification and Traceability

Verification in this public FRD is intentionally black-box. It establishes whether the externally observable requirement has been demonstrated, inspected, or bounded without exposing source code, models, detailed signal mappings, calibration constants, or setup procedures.

Requirement group	Verification basis	Public disposition
Functional behavior	Bench demonstration and retained internal engineering evidence	Demonstrated where stated
External interfaces	Observed system-level communication and feedback behavior	Demonstrated / bounded
Safety and release controls	Document and operational review	Release-controlled

Requirement group	Verification basis	Public disposition
Performance characterization	Summary-level evidence only; detailed methods retained privately	Capability demonstrated; no certified performance claim

Acceptance and Release Criteria

- Functional platform requirements have defined verification methods and public dispositions.
- The public package contains no schematics, BOMs, pinouts, connector maps, board files, harness documentation, or setup/commissioning instructions.
- Third-party product references are descriptive only and do not redistribute vendor design assets.
- Qualification statements remain limited to demonstrated bench capability.

Limitations and Document Control

This document is a public black-box FRD. It is not a design specification, source-code release, manufacturing package, laboratory procedure, or deployment guide. Implementation detail sufficient to reproduce, modify, tune, or manufacture the system is intentionally withheld.

Attribution

Herder Elektronische Systemen prepared this public black-box FRD to document platform function and verification status. Third-party hardware remains subject to its respective ownership and licensing. Detailed physical design, wiring, manufacturing, and commissioning information is not included.